

National University of Computer and Emerging Sciences, Lahore Campus



Course Name: Computer Organization and Assembly Language

Course Code: EE213

Program: BS(Data Science)

Semester: Fall2022

Student : Name: _____ Roll No. _____ Section: _____

Homework# 1

Deadline: 12 September 2022

Submit Handwritten homework in class and also submit scanned file in google classroom.

Late submissions not allowed.

Q1 to Q5 carry 5 marks each.

- Q1. To address 1-MB of memory we need 20 bits of addressing, then how much megabytes of memory can be accessed using 30 bits of addressing. _____ MB.
- Q2. Given the following jump statement and its opcode, identify the type of jump (near or short) and the offset (logical address in CS) to which the jump will take place.

Offset of Opcode	Opcode	Type of Jump?	Offset?
0125	EBE9 ;EB is opcode of jump		

- Q3. Given the following sequential set of instructions of same program, write down the values of CF, PF, ZF after each instruction: (initially all flags are zero.)

	CF	PF	ZF
xor ah, ah	_____	_____	_____
mov al, 0x4A	_____	_____	_____
shl al, 2	_____	_____	_____
rcr ah, 3	_____	_____	_____
sub ah, al	_____	_____	_____

- Q4. Write an instruction which reads the first byte of its own opcode and stores it into al register.

- Q5. Write assembly code to compare two 32-bit numbers such that if num1 is equal to num2 it sets ZF=1 else ZF=0. (Declare two 32-bit numbers in memory to compare)

Q6. Write an assembly program, such that given an array of **ten** integers, your program finds and stores the sum of unique elements of array in a memory label called **“sum”** (defined word). **[10 Marks]**

